A negative answer to polynomial sample complexity for deletion-trace reconstruction

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October 12, 2026

Abstract

Can every binary string be recovered from polynomially many independent deletion traces at a fixed deletion probability? The September 2026 lower bound of OpenAI answers this question negatively. We record its precise consequence for AIM problem 321, including randomized decoders with unrestricted computation and constants depending on the channel. The argument combines the cited one-trace indistinguishability theorem with an elementary product bound and two-point testing. The underlying indistinguishability construction is due to OpenAI.

1 The model and the source result

Fix a known deletion probability $q \in (0,1)$ and an integer $n \geq 2$. For $x \in \{0,1\}^n$, the law $D_q(x)$ deletes each bit independently with probability q and lists the surviving bits in their original order. The output space is

$$\mathcal{T}_n = \bigcup_{j=0}^{n} \{0,1\}^j,$$

where the empty word is included. A trace reveals its length and its bits, but no original positions. The sample is a tuple of independent full traces. A decoder knows n, q, may be randomized, and outputs a length-n binary word. It must be the same decoder for every unknown input.

Problem 321 of the AIM catalogue [1] asks whether, for every fixed q, there are finite $C_q, k_q > 0$ giving exact recovery with probability at least $2/3$ on every input, at every length $n \geq 2$, from at most $\lceil C_q n^{k_q} \rceil$ traces. Computation time is unrestricted. This note explains why the later result of OpenAI [2] rules out that guarantee. It does not introduce a new indistinguishability construction.

For laws P, Q on a finite set S, use the convention

$$\mathrm{TV}(P,Q) = \frac{1}{2} \sum_{z \in S} |P(z) - Q(z)|.$$

The complete trace laws are supported on $\mathcal{T}_n$, so this agrees with the source's sum over all finite binary words. Define

$$\delta_n(q) = \min_{\substack{x,y \in \{0,1\}^n \\ x \neq y}} \mathrm{TV}(D_q(x), D_q(y)).$$

The minimum exists because the collection of distinct pairs is finite and nonempty.

Source theorem (OpenAI, one-trace conclusion of Theorem 1.2)**.** *For a channel with any fixed $q \in (0,1)$ and any real exponent $a > 0$,*

$$\lim_{n \to \infty} n^a \delta_n(q) = 0. \tag{1}$$

The cited statement is equation (1.2) on page 2 of [2]. Its inputs are deterministic binary words, and its observations are complete traces. The limit runs through all integer lengths; Section 8 of that paper proves the bound for complete trace laws and pads the constructed pairs to length n. In particular, q remains fixed as n grows. The proof below uses only (1) from that paper.

2 From indistinguishability to reconstruction

Write $P^{\otimes m}$ for the joint law of m independent samples from P. For $m = 0$ the observation space is a singleton.

Lemma 1. *Suppose two distinct inputs x, y give observation laws P, Q on the same finite set. For every common randomized decoder A,*

$$\frac{\mathbb{P}_x(A = x) + \mathbb{P}_y(A = y)}{2} \le \frac{1 + \mathrm{TV}(P, Q)}{2}. \tag{2}$$

For each nonnegative integer m,

$$\mathrm{TV}(P^{\otimes m}, Q^{\otimes m}) \le m\,\mathrm{TV}(P, Q). \tag{3}$$

Proof. Let $f(z)$ and $g(z)$ be the decoder's conditional probabilities of reporting x and y on observation z. They satisfy $0 \le f, g \le 1$ and $f + g \le 1$. Consequently,

$$\mathbb{P}_x(A = x) + \mathbb{P}_y(A = y) \le \mathbb{E}_P f + \mathbb{E}_Q(1 - f) = 1 + \sum_z (P(z) - Q(z)) f(z).$$

Because $P - Q$ has total mass zero, the sum of its positive entries is $\mathrm{TV}(P, Q)$. The last sum is at most that positive mass, which proves (2). This calculation already averages all randomness used by the decoder.

For $m \ge 1$, the signed mass functions satisfy the exact identity

$$P^{\otimes m} - Q^{\otimes m} = \sum_{j=0}^{m-1} P^{\otimes j} \otimes (P - Q) \otimes Q^{\otimes (m-1-j)}.$$

Each summand has ℓ^1 norm $\|P - Q\|_1$: the other factors are probability laws with mass one. The triangle inequality and division by two give (3). For $m = 0$ both product laws are the same point mass. □

Theorem 1. *For every fixed $q \in (0, 1)$ and all finite real constants $C, k > 0$, there is an integer $n_0 \ge 2$ such that the following holds. For every integer $n \ge n_0$, every integer $0 \le m \le \lceil Cn^k \rceil$, and every randomized decoder $A : \mathcal{T}_n^m \to \{0,1\}^n$, some input $x \in \{0,1\}^n$ satisfies*

$$\mathbb{P}[A(Y_1, \ldots, Y_m) = x] < \frac{2}{3}, \qquad Y_1, \ldots, Y_m \overset{\text{iid}}{\sim} D_q(x).$$

Thus the polynomial-sample guarantee in AIM problem 321 fails at every fixed deletion probability in $(0, 1)$.

Here n is the known input length, m is the number of independent complete traces, and the probability includes the decoder's own randomness. The threshold n_0 may depend on q, C, k; no uniformity in the channel is required.

Proof. Fix q, C, k as stated. At each $n \geq 2$ choose distinct x_n, y_n attaining $\delta_n(q)$. They can be chosen before the sample budget and the decoder. Since $n^k \geq 1$, every permitted budget obeys

$$m \leq \lceil Cn^k \rceil \leq Cn^k + 1 \leq (C+1)n^k.$$

Lemma 1 therefore gives

$$\mathrm{TV}(D_q(x_n)^{\otimes m}, D_q(y_n)^{\otimes m}) \leq (C+1)n^k \delta_n(q) =: \varepsilon_n.$$

Equation (1) with $a = k$ implies $\varepsilon_n \to 0$ at this same fixed channel. Choose $n_0 \geq 2$ so that $\varepsilon_n < 1/3$ for every integer $n \geq n_0$. Applying (2) to the full m-trace laws,

$$\frac{\mathbb{P}_{x_n}(A = x_n) + \mathbb{P}_{y_n}(A = y_n)}{2} \leq \frac{1 + \varepsilon_n}{2} < \frac{2}{3}.$$

At least one of these two recovery probabilities is strictly below 2/3. The pair works for every permitted m and every common decoder. This proves the stated quantifiers and contradicts any polynomial guarantee valid at all lengths. □

AI-use statement. OpenAI Codex assisted with source retrieval, drafting, and checking the testing argument in this note. The multiscale lower-bound theorem is credited to OpenAI [2]. Independent mathematical review is requested.

References

[1] AIM catalogue, *321. Polynomial sample complexity for deletion-trace reconstruction*, repository version `c9929805cece044705abaaebe1cebddac1ba11a6`. https://github.com/MColbrook/AIM/blob/c9929805cece044705abaaebe1cebddac1ba11a6/problems/321-polynomial-trace-reconstruction.md.

[2] OpenAI, *Quantitative lower bounds for trace reconstruction*, OpenAI Math Release preprint, September 24, 2026. Theorem 1.2, p. 2, equation (1.2); proof in Section 8 following the construction in Sections 2–7. Repository version `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. https://github.com/openai/math/blob/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/preprints/quantitative-lower-bounds-for-trace-reconstruction-September-24-2026/paper.pdf.